

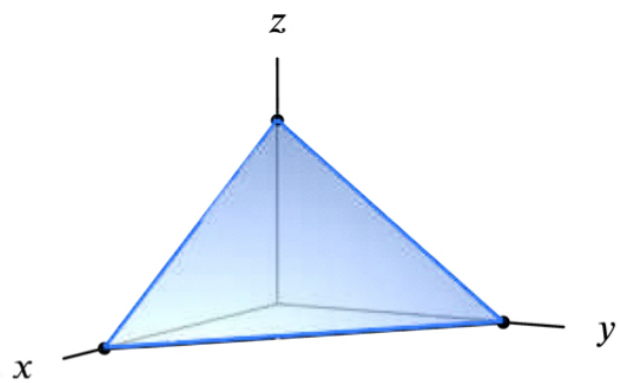
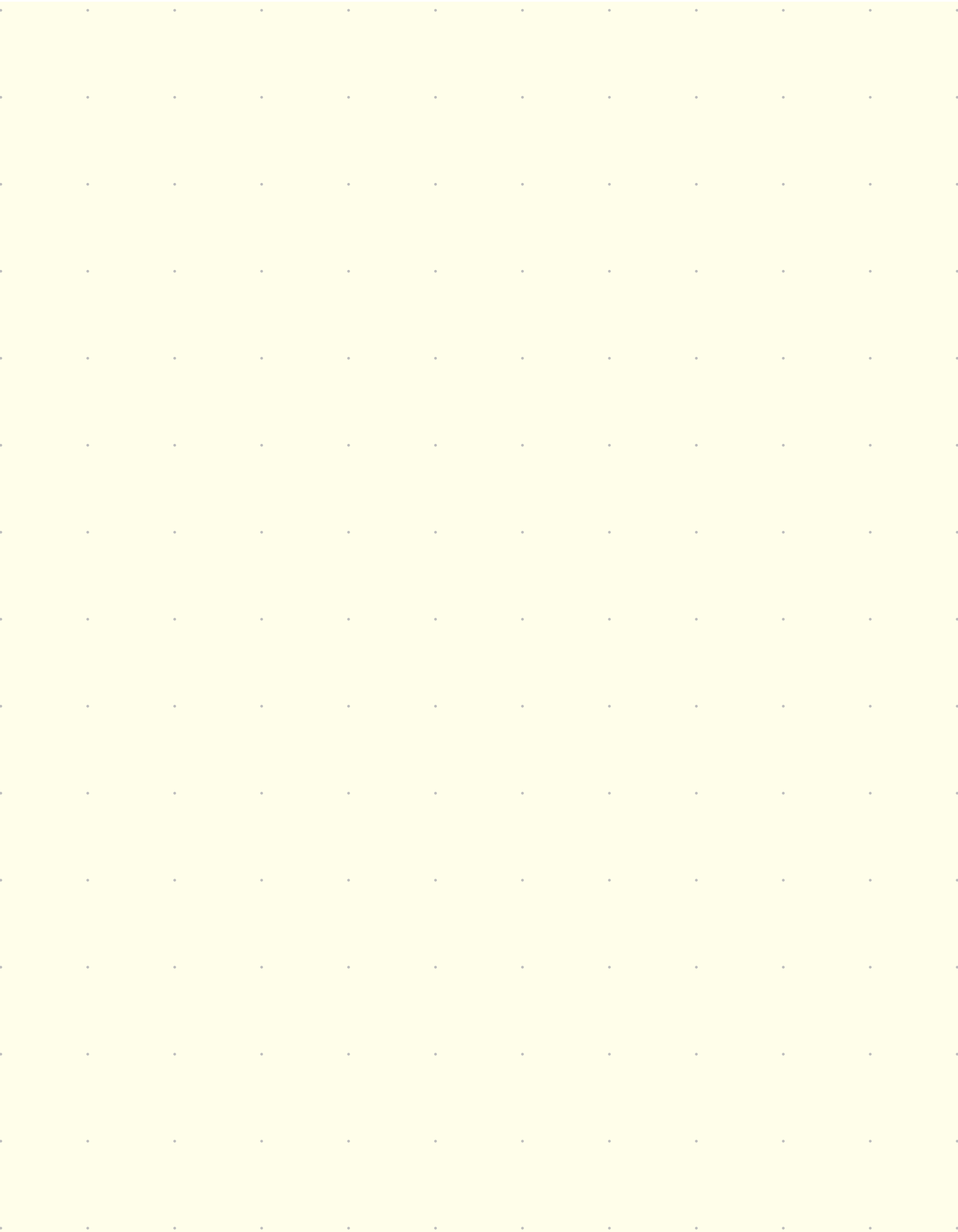


Figure 12.79

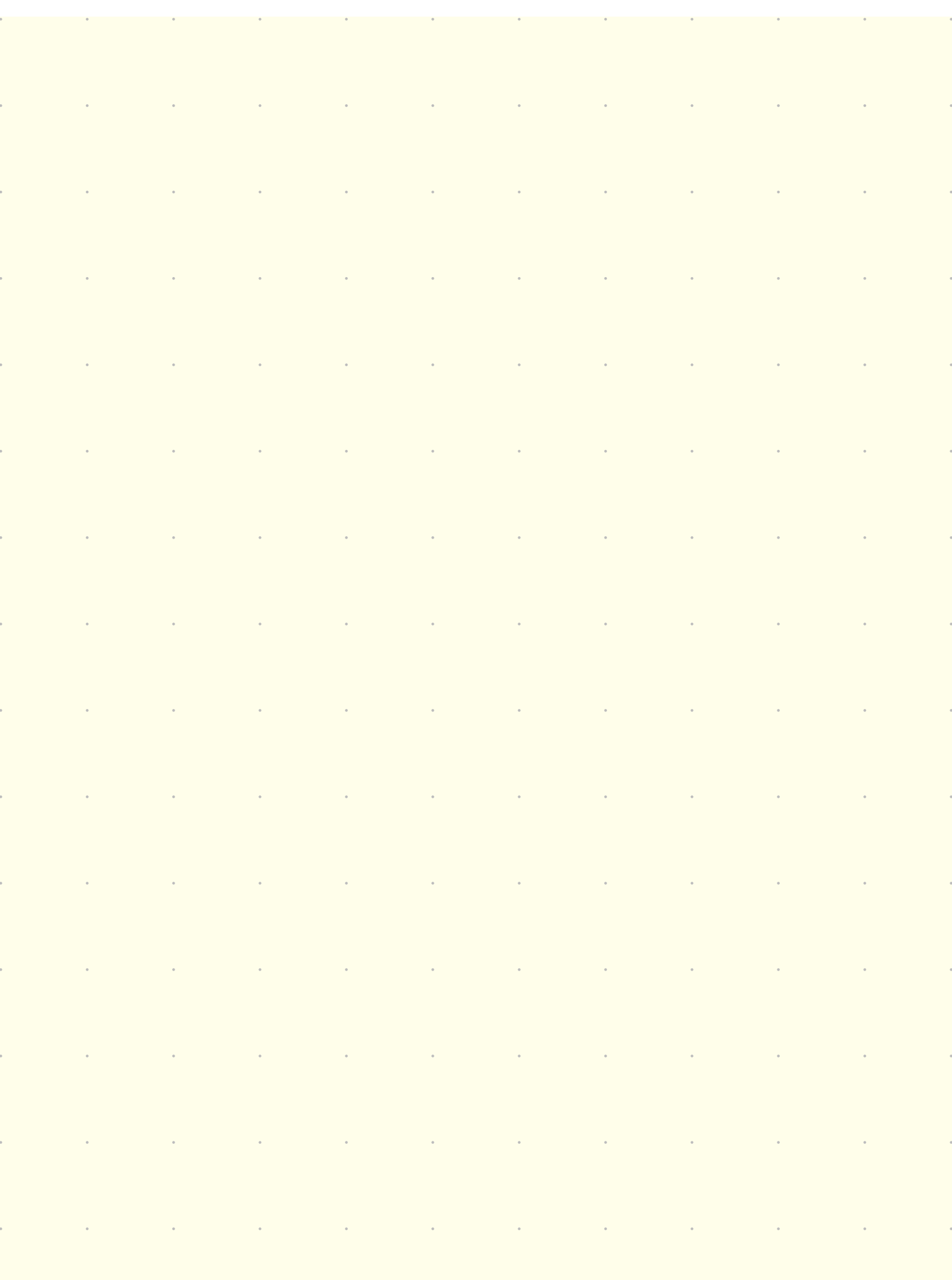
39. $z = 2 - 2x + y$

40. $z = 2 - x - 2y$

Preliminary Example A. Find an equation for a sphere of radius r centered at the origin.



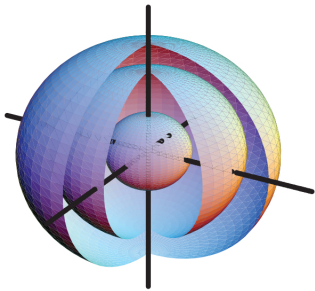
Preliminary Example B. Sketch graphs of the following three surfaces in three-dimensional space:
 $x^2 + y^2 = 4$, $x^2 + z^2 = 9$, and $y = x^2$.



Example 2. A block of ice is located at the point $(0,0,0)$. The temperature at the point (x,y,z) is given by $T(x,y,z) = \frac{1}{4}(x^2 + y^2 + z^2)$ degrees Celsius, where x,y and z are measured in meters.

(a) How many dimensions would be required to represent the graph of T ?

(b) Pictured to the right are the surfaces $T(x,y,z) = 1$, $T(x,y,z) = 5$, and $T(x,y,z) = 9$. Find formulas for these surfaces and explain their significance in the context of this problem.



A large grid of dots for writing the answer to part (b).

Describe the level surfaces of each of the following functions.

(a) $f(x, y, z) = e^{-(x^2+y^2)}$

(b) $g(x, y, z) = z - y$

(c) $V(x, y, z) = \ln(x^2 + y^2 + z^2)$

